Data Acquisition & Market Sources

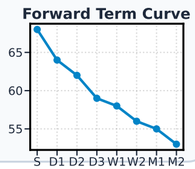
Physical Grid Fundamentals (REE)

- 16 Continuous grid features (Hourly to daily)
- Generation mix: Wind, Solar, Hydro, CC
- Grid demand & generation imbalance



OMIP Derivatives & OMIE Spot

- 14 Tradable contracts (Baseload SPEL)
- Futures: D1–D3, W1–W4, M1–M6
- OMIClear daily margin clearing



Macroeconomic News Feeds

- 1,242 Consecutive market days (2023–2026)
- elEconomista, Europa Press, OMIE/OMIP/REE
- Gas supply, EU ETS carbon & geopolitics

LLM News Sentiment & Feature Engineering

1. Zero-Shot LLM Sentiment (Llama-3 8B)

- Local Ollama inference (Zero internet / No data leakage)
- Numeric structured sentiment extraction in [-1.0, +1.0]

φ_1-φ_3: Short/Med/Long Outlook	φ_4: LNG & Natural Gas
φ_5: EU ETS CO_2 Policy	φ_6: MIBEL Regulation
φ_7: Generation Mix / Hydro	φ_8-φ_{10}: Geopolitics & Macro

2. Time-Aware Feature Engineering Pipeline

1. **1. Strict Temporal Lag Enforcement**
   Features conditioned strictly on data published up to t-1.
2. **2. Temporal Imputation**
   Forward-filling weekend market quotes with NO_TRADE masks.
3. **3. Rolling Standardization**
   Z-score scalers fit strictly on historical training slice D_train.
4. **4. Exploratory Factor Structure**
   Confirms sentiment signals add non-collinear variance to grid.

Feature Representations & Factorial Ablation

Strict 2×2 Factorial Ablation

Base (X_Common)
Dataset_Exogenous_Calendar
- Physical + OMIP + Calendar
- Dim: 35 features

Quant (X_Quant)
Dataset_Exogenous_Commodities
- Common (35) + 7 Commodities
- Dim: 42 features

Hybrid (X_Hybrid)
Dataset_M8_Physical_NLP
- Common (35) + 10 Sentiment
- Dim: 45 features

Full (X_Full)
Dataset_NLP_Subgroups
- Common + Commodities + NLP
- Dim: 52 features

Controlled Comparisons

- ▸ **NLP over Quant:** X_Full vs X_Quant (+1.59 pp in 2025).
- ▸ **NLP over Base:** X_Hybrid vs X_Common (+4.18 pp in 2025).
- ▸ **Text vs Commodities:** X_Hybrid outperforms in 2026 shock.
- ▸ **Noise Robustness:** Tested vs uncurated news dataset.

DRL Model Formulation & Walk-Forward Retraining

1. Markov Decision Process (MDP) Procurement

- Action Space A: 14 Discrete Contracts (SPOT, D1–D3, W1–W4, M1–M6)
- Observation o_t: 7-day lagged state + 400-day forward inventory queue c_t
- Reward Function: $r_t = (P_{SPOT,t+1} - P_{eff,t+1}) - R_{penalty} \cdot I_{invalid}$
- Clearing Rules: OMIClear variation margins + OMIE physical delivery

2. Deep Recurrent & Attention Architectures

RecurrentPPO (LSTM)
- Hidden state h_t (d_h = 256 / 64)
- Generalized Advantage Estimation (GAE)

GTrXL Transformer
- Multi-Head Attention (H=4, L=2, M=14)
- GRU skip connections & LayerNorm

3. Continuous Walk-Forward Protocol (B = 41 Blocks)

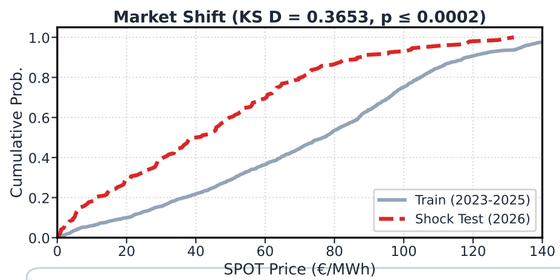
- Rolling Update Horizons W in {1, 3, 6, 12} months across 2023–2026
- Bounded compute budget (K = 15,000 steps/month; warm-start updates)
- Multi-Seed Robustness: 5 independent evaluation seeds (s = 10..50)

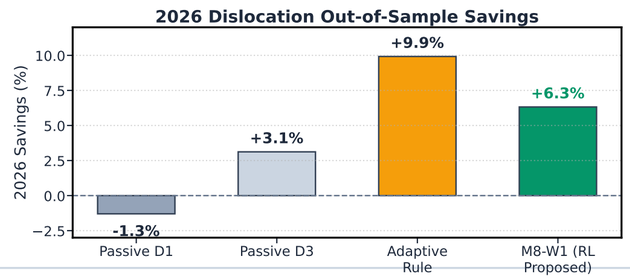
Train: M_{b-1} (W) → Out-of-Sample: M_b → Persist Inventory c_{eval} → M_{b+1}

Performance & Downside Risk Mitigation (2026 Dislocation)

1. Core Empirical Findings & Downside Risk Protection

- +6.32% Multi-Seed Savings under 2026 wholesale deflation (SPOT €44.22/MWh).
- Mitigated Maximum Drawdown: Reduced by ~66% vs D3 (186.45 vs 548.95 €/MWh).
- Lower Downside Risk: $\sigma_- = 9.15$ €/MWh (vs 13.68 for D2 and 18.63 for D3).
- Dynamic Multi-Horizon Allocation: Hedging across D2, D1, SPOT, and D3.





2. SHAP Explainability & Macroeconomic Grounding

- Feature attribution confirms policy systematically attends to news sentiment.
- Regulatory Risk (φ_6), LNG (φ_4), and Generation Mix (φ_7) act as leading signals.
- RL dynamically avoids locking overpriced long-horizon contracts in down-markets.
- Sensitivity analysis under linear transaction fees (up to 0.25 €/MWh) & 5 seeds.

